Set 6 Problem 5: Let $G \subseteq \mathbb{C}$ be a simply connected region with $G \neq \mathbb{C}$.

(a) Show that any holomorphic map $f : G \to G$ other than the identity admits at most one fixed point.

Suppose for the sake of contradiction that f has two distinct fixed points $a, b \in G$. By the Riemann mapping theorem, we know there is a biholomorphism $\theta : G \to \Delta$. In turn, we know that $g := \theta \circ f \circ \theta^{-1}$ is a holomorphic function from Δ to Δ. Also, $g := \theta \circ f \circ \theta^{-1}$ has two distinct fixed points $\alpha := \theta(a)$ and $\beta := \theta(b)$. If we can prove that g is the identity map, then we will also know that f is the identity map.

Given any $z_0 \in \Delta$, let $\varphi_{z_0}(z)$ denote the biholomorphism of the form $\frac{z-z_0}{1-\overline{z_0}z}$. Then note that $h := \varphi_\alpha \circ g \circ \varphi_{-\alpha}$ is a holomorphism of Δ with the distinct fixed points 0 and $\varphi_\alpha(\beta)$. By Schwarz's lemma, we can thus conclude that h is a rotation. Yet the only rotation with a fixed point besides 0 is the identity map. So, h is the identity map. In turn, so is g.

(b) Show that the above statement is false for $G = \mathbb{C}$.

Let $f(z) = z^2$. Then f is a holomorphic map from $\mathbb{C}$ to $\mathbb{C}$ with 0 and 1 as fixed points. This contradicts part (a).

(c) Now assume $f : G \to G$ is holomorphic and admits a fixed point a such that $f'(a) = 1$. Show that f must be the identity.

Let $\theta : G \to \Delta$ be a biholomorphic function sending a to 0. Then note that $g = \theta \circ f \circ \theta^{-1}$ is a holomorphic function from Δ to Δ such that $g(0) = 0$. Also note that:

$$\begin{aligned} g'(0) &= \theta'(f(\theta^{-1}(0))) \cdot f'(\theta^{-1}(0)) \cdot (\theta^{-1})'(0) \\ &= \theta'(a) \cdot f'(a) \cdot (\theta^{-1})'(0) = \theta'(a) \cdot f'(a) \cdot \frac{1}{\theta'(\theta^{-1}(0))} = \frac{\theta'(a)}{\theta'(a)} \cdot f'(a) = 1 \end{aligned}$$

It follows by Schwarz' lemma that g is a rotation. Also, since the only rotation with $g'(0) = 1$ is the identity map, we know that g is the identity map. This in turn proves that f is also the identity map. ■

Math 200b Notes:

Before getting into more notes, I want to do a bunch of book-keeping.

- I already mentioned that if $F, G : \mathbf{D} \to \mathbf{E}$ and $H : \mathbf{C} \to \mathbf{D}$ are functors with $\eta : F \to G$ being a natural transformation / isomorphism, then we can define a natural transformation / isomorphism $\theta : F \circ H \to G \circ H$ by setting $\theta_a = \eta_{H(a)}$ for all $a \in \mathrm{Obj}(\mathbf{C})$.

  In a similar vein, note that if $F : \mathbf{D} \to \mathbf{E}$ and $G, H : \mathbf{C} \to \mathbf{D}$ are functors with $\eta : G \to H$ being a natural transformation / isomorphism, then we can define a natural transformation / isomorphism $\theta : F \circ G \to F \circ H$ by setting $\theta_a = F(\eta_a)$ for all $a \in \mathrm{Obj}(\mathbf{C})$.

  A key fact I'm implicitely using here is that if $f : a \to b$ is an isomorphism then $F(f) : F(a) \to F(b)$ is also an isomorphism for any functor F with an inverse

given by $F(f^{-1})$. This is a result of the fact that functors preserve compositions of morphisms.

- As I start to talk more about multi-argument functors (i.e. functors from products of categories), it will be more and more helpful to talk about contravariant functors $F : \mathbf{C} \to \mathbf{D}$ as instead being ordinary functors from $\mathbf{C}^{\mathrm{op}}$ to $\mathbf{D}$. However, there is a downside to this approach that I want to take note of.

  Suppose $F : \mathbf{C} \to \mathbf{D}$ and $G : \mathbf{D} \to \mathbf{E}$ are contravariant functors. Then the codomain of F and the domain of G are the same and hence $G \circ F$ gives a well-defined mapping of all morphisms and objects (which we can easily check to be a functor).

  That said, if we instead view F and G as ordinary functors from opposite categories, then the codomain of F (which is $\mathbf{D}$) and the domain of G (which is $\mathbf{D}^{\mathrm{op}}$) are no longer the same categories.

  Fortunately, note that for any functor $F : \mathbf{C} \to \mathbf{D}$ we can define a functor $\overline{F} : \mathbf{C}^{\mathrm{op}} \to \mathbf{D}^{\mathrm{op}}$ given by $\overline{F}(a) = F(a)$ and $\overline{F}(f^{\mathrm{op}}) = (F(f))^{\mathrm{op}}$. Then by an abuse of notation, if $G : \mathbf{D}^{\mathrm{op}} \to \mathbf{E}$ is another functor, we will write $G \circ F$ to denote $G \circ \overline{F}$.

  (Also note that $\overline{\overline{F}} = F$...)

- Next, I want to finally make sense of parts (b) and (c) of the homework problem on problem on *pages 683-684*. A brief note on notation is that $\mathrm{Fun}(\mathbf{C}, \mathbf{D})$ is just another way of denoting the category of functors from $\mathbf{C}$ to $\mathbf{D}$.

  Note that part (b) defines a functor $\mathbf{C}^{\mathrm{op}} \to \mathrm{Fun}(\mathbf{C}, \mathbf{Set})$ given by

$$\begin{tikzcd} a \arrow[d, "f^{\mathrm{op}}"] & h_a \arrow[d, "\widehat{f}"'] \\ a' & h_{a'} \end{tikzcd}$$

  where $\widehat{f} : h_a \to h'_a$ is the natural transformation given by $\widehat{f}_b(a \xrightarrow{g} b) := a' \xrightarrow{g \cdot f} b$ for all $b \in \mathrm{Obj}(\mathbf{C})$.

  Technically in that homework problem I did not show that composition of morphisms behaves properly. This is largely because at that time I hadn't yet defined the category $\mathrm{Fun}(\mathbf{C}, \mathbf{Set})$. But now that I have defined that category, this is easy to show.

  - For all $a, b \in \mathrm{Obj}(\mathbf{C})$ we have that $(\widehat{1_a})_b(a \xrightarrow{g} b) = a \xrightarrow{g \cdot 1_a} b = \mathrm{Id}_{h_a(b)}$. Therefore our proposed functor maps identity morphisms of $\mathbf{C}$ to identity natural transformations between functors (i.e. identity morphisms of $\mathrm{Fun}(\mathbf{C}, \mathbf{Set})$).

  - Suppose $f_1^{\mathrm{op}} : a \to a'$ and $f_2^{\mathrm{op}} : a' \to a''$ are morphisms in $\mathbf{C}$. Then for any morphism $g \in h_a(b)$ and $b \in \mathrm{Obj}(\mathbf{C})$, we have that:

$$(f_1 \cdot f_2)^{\wedge}_b(g) = g \cdot f_1 \cdot f_2 = (\widehat{f_2})_b(g \cdot f_1) = ((\widehat{f_2})_b \circ (\widehat{f_1})_b)(g)$$

And this shows that $(f_1 \cdot f_2)^\wedge_b(g) = (\widehat{f_2})_b \circ (\widehat{f_1})_b$ (which is what we wanted).

Letting the prior functor $\mathbf{C}^{\mathrm{op}} \to \mathrm{Fun}(\mathbf{C}, \mathbf{Set})$ be denoted as H, part (c) tell us that the isomorphisms defined in part (a) induce a natural isomorphism between the functors $h^F \circ H$ and F.

- On a related note, I want to briefly return to the comment at the top of *page 708*. Suppose $\mathbf{C}$ is a locally finite category and $F_1, F_2 : \mathbf{C} \to \mathbf{C}$ are functors. Then for each i we can define a functor $H_i : \mathbf{C} \to \mathrm{Fun}(\mathbf{C}^{\mathrm{op}} \to \mathbf{Set})$ by:

$$\begin{tikzcd} x \arrow[d, "f"] & & h^x \circ F_i \arrow[d, "\widehat{f}"] \\ x' \arrow[rr, squiggly, "H_i"] & & h^{x'} \circ F_i \end{tikzcd}$$

where $\widehat{f}$ is the natural transformation given by $\widehat{f}_y(F(y) \xrightarrow{h} x) := F(y) \xrightarrow{f \cdot h} x'$ for all $y \in \mathrm{Obj}(\mathbf{C})$

Note that this is a well-defined natural transformation since for any morphism $g^{\mathrm{op}} : y \to y'$ in C^{op} we have that:

$$\begin{tikzcd} h \arrow[rrr, mapsto] \arrow[dddd, mapsto] & & & f \cdot h \arrow[dddd, mapsto] \\ & h^x \circ F_i(y) \arrow[r, "\widehat{f}_y"] \arrow[d, "h^x(F_i(g))"] & h^{x'} \circ F_i(y) \arrow[d, "h^{x'}(F_i(g))"'] & \\ & h^x \circ F_i(y') \arrow[r, "\widehat{f}_{y'}"'] & h^{x'} \circ F_i(y') & \\ h \cdot F_i(g) \arrow[rrr, mapsto] & & & f \cdot h \cdot F_i(g) \end{tikzcd}$$

Note that in the problem on *pages 705-708*, $G_i(x, y) = (H_i(x))(y)$ and:

$$G_i((f, g^{\mathrm{op}}) : (x, y) \to (x', y')) = (H_i(x'))(g^{\mathrm{op}}) \circ (H_i(f))_y = (H_i(f))_{y'} \circ (H_i(x))(g^{\mathrm{op}})$$

With this is mind, one can check that the statement "$h^x \circ F_1 \cong h^x \circ F_2$ naturally in x" is equivalently referring to the functors H_1 and H_2 being naturally isomorphic. I won't actually write a proof though.

Notice a natural transformation $\eta : H_1 \to H_2$ will consist of a class of natural transformations $\eta_x : h^x \circ F_1 \to h^x \circ F_2$ for each $x \in \mathrm{Obj}(\mathbf{C})$ such that the below diagram commutes in the category $\mathrm{Fun}(\mathbf{C}^{\mathrm{op}}, \mathbf{Set})$ for every morphism $f : x \to x'$ in $\mathbf{C}$:

$$\begin{tikzcd} h^x \circ F_1 \arrow[r, "\eta_x"] \arrow[d, "\widehat{f}"'] & h^x \circ F_2 \arrow[d, "\widehat{f}"'] \\ h^{x'} \circ F_1 \arrow[r, "\eta_{x'}"] & h^{x'} \circ F_2 \end{tikzcd}$$

In particular, the above diagram commutes iff $\widehat{f} \circ \eta_{x,y} = \eta_{x',y} \circ \widehat{f}$ for all $y \in \mathrm{Obj}(\mathbf{C})$. Also η is a natural isomorphism iff each η_x is a natural isomorphism.

I'll finally discuss things from class now.

Given two functors $L : \mathbf{C} \to \mathbf{D}$ and $R : \mathbf{D} \to \mathbf{C}$ going between locally small categories, we say that L and R are <u>left and right adjoints</u> respectively if for all $c \in \mathbf{C}$ and $d \in \mathbf{D}$ there exists a bijective function $\eta_{(d,c)}$ such that for all $f \in \mathrm{Hom}_{\mathbf{D}}(d_1, d_2)$ and $g^{\mathrm{op}} \in \mathrm{Hom}_{\mathbf{C}^{\mathrm{op}}}(c_1, c_2)$ we have the following commuting diagram:

$$\begin{tikzcd}
h \arrow[rrr, mapsto] \arrow[dd, mapsto] & & & \eta_{(d_1,c_1)}(h) \arrow[dd, mapsto] \\
& \mathrm{Hom}_{\mathbf{D}}(L(c_1), d_1) \arrow[r, "\eta_{(d_1,c_1)}"] \arrow[d] & \mathrm{Hom}_{\mathbf{C}}(c_1, R(d_1)) \arrow[d] & \\
& \mathrm{Hom}_{\mathbf{D}}(L(c_2), d_2) \arrow[r, "\eta_{(d_2,c_2)}"] & \mathrm{Hom}_{\mathbf{C}}(c_2, R(d_2)) & R(f)\cdot\eta_{(d_1,c_1)}(h)\cdot g \arrow[d, equal] \\
f\cdot h\cdot L(g) \arrow[rr, mapsto] & & & \eta_{(d_2,c_2)}(f\cdot h\cdot L(g))
\end{tikzcd}$$

For an example of left and right adjoints, recall that $h_{M_2} \circ h_{M_1} \cong h_{M_1 \otimes M_2}$ by the natural isomorphisms induced by $\eta_{(N,M_1,M_2)}$:

$$\begin{tikzcd}
\mathrm{Hom}_A(T_{M_1}(M_2), N) \arrow[r, "\sim"] & \mathrm{Hom}_A(M_2, h_{M_1}(N)) \\
f{:}M_1 \otimes M_2 \to N \arrow[r, mapsto] & \phi{:}M_2 \to \mathrm{Hom}_A(M_1, N) \\
& (\phi(m_2))(m_1) = f(m_1 \otimes m_2)
\end{tikzcd}$$

In fact though, we can actually go a step further and define functors

$$H_1, H_2 : (A\text{-}\mathbf{mod}) \times (A\text{-}\mathbf{mod})^{\mathrm{op}} \times (A\text{-}\mathbf{mod})^{\mathrm{op}} \to A\text{-}\mathbf{mod} \text{ by:}$$

$$\begin{tikzcd}
N \arrow[r, "f"] & N' & & h \arrow[dd, mapsto] & h^N \circ T_{M_1}(M_2) \arrow[dd, "H_1((f, g_1^{\mathrm{op}}, g_2^{\mathrm{op}}))"] \\
M_1 \arrow[r, "g_1^{\mathrm{op}}"] & M_1' \arrow[rrr, squiggly, "H_1"] & & & \\
M_2 \arrow[r, "g_2^{\mathrm{op}}"] & M_2' & & & h^{N'} \circ T_{M_1'}(M_2') \\
& & & f\circ h\circ(g_1 \otimes g_2) &
\end{tikzcd}$$

and:

$$\begin{tikzcd}
N \arrow[r, "f"] & N' & & (\phi(m_2))(m_1) \arrow[dd, mapsto] & h_{M_2}(h_{M_1}(N)) \arrow[dd, "H_2((f, g_1^{\mathrm{op}}, g_2^{\mathrm{op}}))"] \\
M_1 \arrow[r, "g_1^{\mathrm{op}}"] & M_1' \arrow[rrr, squiggly, "H_2"] & & & \\
M_2 \arrow[r, "g_2^{\mathrm{op}}"] & M_2' & & & h_{M_2'}(h_{M_1'}(N')) \\
& & & (\overline{\phi}(m_2'))(m_1') = (f\circ(\phi\circ g_2)(m_2')\circ g_1)(m_1') &
\end{tikzcd}$$

I do not feel like proving that H_1, H_2 are well-defined functors. Anyways, it's easy to see that the $\eta_{(N,M_1,M_2)}$ define a natural isomorphism between H_1 and H_2. In the special case that we fix M_1 and set $g_1^{\text{op}} = \text{Id}_{M_1}^{\text{op}}$, then this proves that T_{M_1} and h_{M_1} are left and right adjoints of each other for all fixed M_1.

Proposition: If $L, R : A$-**mod** $\to A$-**mod** are left and right adjoints respectively of each other, then L is right-exact and R is left-exact.

Proof:

Let $\{0\} \longrightarrow M_1 \xhookrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \longrightarrow \{0\}$ be an SES. Then we want to show that the following are both SES's:

$$\begin{tikzcd} \{0\} \arrow[r] & R(M_1) \arrow[r, hook, "R(f_1)"] & R(M_2) \arrow[r, "R(f_2)"] & R(M_3) & \\ & L(M_1) \arrow[r, "L(f_1)"] & L(M_2) \arrow[r, two heads, "L(f_2)"] & L(M_3) \arrow[r] & \{0\} \end{tikzcd}$$

Fortunately, by the proposition on *pages 638-640* we know that it is enough to argue for every A-module N that the following sequences are exact:

$$\begin{tikzcd} \{0\} \arrow[r] & h_N \circ R(M_1) \arrow[r, hook, "h_N \circ R(f_1)"] & h_N \circ R(M_2) \arrow[r, "h_N \circ R(f_2)"] & h_N \circ R(M_3) & \\ & h^N \circ L(M_1) & h^N \circ L(M_2) \arrow[l, "h^N \circ L(f_1)"'] \arrow[r, hook, "h^N \circ L(f_2)"] & h^N \circ L(M_3) \arrow[l] & \{0\} \arrow[l] \end{tikzcd}$$

Yet by the definition of a left and right adjoint, we know that $h_N \circ R(M) \cong h_{L(N)}(M)$ and $h^N \circ L(M) \cong h^{R(N)}(M)$ naturally (in both N and M). And since $h_{L(N)}$ and $h^{R(N)}$ are both left exact functors for all N, this proves that the above sequences are exact. ■

Corollary: The tensor functor T_M is right-exact.

Now one may ask about when T_M is fully exact. We know it is possible for this to happen since we already proved on *page 701* that T_A is naturally isomorphic to the identity functor on the category of A-modules and is in turn an exact functor.

We say M is a flat A-module if T_M is an exact functor.

Observe that T_M is an exact functor iff $T_M(f) = \text{Id}_M \otimes f$ is injective when f is injective.

Before going into some theorems on flat modules, we'll need to prove some other results.

Lemma: $T_M \circ T_{M'} \cong T_{M \otimes M'}$.

Proof:

By set 7 problem 3 (on *pages 705-708*), it suffices to prove that:

$$h^N \circ (T_M \circ T_{M'}) \cong h^N \circ T_{M \otimes M'} \text{ naturally in } N.$$

Fortunately, we already know that $h^N \circ T_M \cong h^{h_M(N)}$ naturally in N. In turn, by composing functors into the above identity, we can conclude that:

$$h^N \circ (T_M \circ T_{M'}) \cong h^{h_M(N)} \circ T_{M'} \cong h^{h_{M'}(h_M(N))} \text{ naturally in } N.$$

Also, we can easily see that $h^{h_{M'}(h_M(N))} \cong h^{h_{M\otimes_A M'}(N)}$ naturally in N. In turn:
$h^{h_{M\otimes_A M'}(N)} \cong h^N \circ T_{M\otimes_A M'}$ naturally in N.

Therefore $h^N \circ (T_M \circ T_{M'}) \cong h^N \circ T_{M\otimes_A M'}$ naturally in N. ■

On a homework problem which I'll later do on *page* ____, I'll give an explicit construction of an isomorphism between $T_M \circ T_{M'}$ and $T_{M\otimes_A M'}$.

<u>Lemma:</u> $T_{\bigoplus_{i\in I} M_i} \cong \bigoplus_{i\in I} T_{M_i}$.

Proof:
Similarly to the prior lemma, we have that:

$$h^N \circ T_{\bigoplus_{i\in I} M_i} \cong h^{h_{\oplus_{i\in I} M_i}} \cong h^{\prod_{i\in I} h_{M_i}(N)}$$
$$\cong \prod_{i\in I} h^{h_{M_i}(N)} \cong \prod_{i\in I}(h^N \circ T_{M_i}) \cong h^N \circ \bigoplus_{i\in I} T_{M_i}$$

naturally in N.

These natural isomorphisms are easy to justify. I don't want to write down the arguments though cause it's late. ■

<u>Lemma:</u> $\bigoplus_{i\in I} F_i$ is exact iff F_i is exact for all $i \in I$.

Hopefully this is obvious.

As an application of the prior lemmas plus that the fact that A is a flat A-module, we can conclude the following results.

1. If $\bigoplus_{i\in I} M_i$ is flat iff $\forall i \in I$ we have that M_i is flat.

   Why: $\bigoplus_{i\in I} M_i$ is flat iff $T_{\bigoplus_{i\in I} M_i}$ is exact, which in turn happens iff $\bigoplus_{i\in I} T_{M_i}$ is exact, and that happens iff for all $i \in I$ we have that T_{M_i} is exact, and finally that happens iff for all $i \in I$ we have that M_i flat.

2. Free A-modules are flat.

3. Any projective A-module is flat.

   Why: If P is projective then there exists an A-module K such that $P \oplus K$ is a free module. Thus $P \oplus K$ is flat, and in turn both P and K are flat.

4. If M_1, M_2 are flat then so is $M_1 \otimes_A M_2$.

   Why: If both M_i are flat then both T_{M_i} are exact. In turn, $T_{M_1} \circ T_{M_2}$ is exact, which means that $T_{M_1\otimes_A M_2}$ is exact. And finally, this implies that $M_1 \otimes_A M_2$ is flat.

Math 200b Homework:

Set 7 Problem 4: Suppose A is a unital commutative ring and $S \subseteq A$ is a multiplicatively closed subset, and M is an A-modules.

(b) Prove that $S^{-1}A \otimes_A M \cong S^{-1}M$ and deduce that $S^{-1}A$ is a flat A-module.

Firstly, note that $S^{-1}A$ is an A-algebra by the identification of a with $\frac{a}{1}$. Therefore, we can enrinch the tensor functor $T_{S^{-1}A}$ to be mapping into $S^{-1}A$-**mod** by considering $S^{-1}A \otimes_A M$ to be an $S^{-1}A$-module where we define $\frac{a}{s}(\frac{a'}{s'} \otimes m) = \frac{aa'}{ss'} \otimes m$.

Define $f : S^{-1}A \times M \to S^{-1}M$ by $f(\frac{a}{s}, m) = \frac{a}{s} \cdot \frac{m}{1}$. This is a well-defined map since we already know that scaling elements of $S^{-1}M$ by elements of $S^{-1}A$ is well-defined. I also claim that it is A-bilinear.

Since $S^{-1}M$ is an $S^{-1}A$-module, we can see that:

$$\begin{aligned} f((\tfrac{a_1}{1} \cdot \tfrac{a}{s}) + (\tfrac{a_2}{1} \cdot \tfrac{a'}{s'}), m) &= \Big((\tfrac{a_1}{1} \cdot \tfrac{a}{s}) + (\tfrac{a_2}{1} \cdot \tfrac{a'}{s'})\Big) \cdot \tfrac{m}{1} \\ &= \tfrac{a_1}{1}(\tfrac{a}{s} \cdot \tfrac{m}{1}) + \tfrac{a_2}{1}(\tfrac{a'}{s'} \cdot \tfrac{m}{1}) = \tfrac{a_1}{1} f(\tfrac{a}{s}, m) + \tfrac{a_2}{1} f(\tfrac{a'}{s'}, m) \end{aligned}$$

Similarly, $f(\frac{a}{s}, a_1 m + a_2 m) = \frac{a}{s} \cdot \frac{a_1 m + a_2 m'}{1} = \frac{a_1}{1} \cdot (\frac{a}{s} \cdot \frac{m}{1}) + \frac{a_2}{1} \cdot (\frac{a}{s} \cdot \frac{m'}{1})$
$= \frac{a_1}{1} f(\frac{a}{s}, m) + \frac{a_2}{1} f(\frac{a}{s}, m)$.

By the universal property of the tensor product, there thus exists a unique A-module homomorphism $\widehat{f} : S^{-1}A \otimes_A M \to S^{-1}M$ such that $\widehat{f}(\frac{a}{s} \otimes m) = \frac{a}{s} \cdot \frac{m}{1}$ for all pure tensor elements $\frac{a}{s} \otimes m$.

Going the other way, define $g : S^{-1}M \to S^{-1}A \otimes_A M$ by $g(\frac{m}{s}) = \frac{1}{s} \otimes m$. To see that this is well-defined, suppose $\frac{m}{s} = \frac{m'}{s'}$. Then there exists $t \in S$ such that $t(s'm - sm') = 0$. In turn, by spamming the A-balance property we get that:

$$\begin{aligned} (\tfrac{1}{s} \otimes m) - (\tfrac{1}{s'} \otimes m) &= (\tfrac{1}{tss'} \otimes ts'm) - (\tfrac{1}{tss'} \otimes tsm') \\ &= \tfrac{1}{tss'} \otimes (ts'm - tsm') = \tfrac{1}{tss'} \otimes 0 = 0 \end{aligned}$$

I also claim that g is $S^{-1}A$-linear. After all:

$$\begin{aligned} g((\tfrac{a_1}{s_1} \cdot \tfrac{m}{s}) + (\tfrac{a_2}{s_2} \cdot \tfrac{m'}{s'})) &= g(\tfrac{a_1 s' s_2 m + a_2 s s_1 m'}{s_1 s \cdot s_2 s'}) \\ &= \tfrac{1}{s_1 s \cdot s_2 s'} \otimes (a_1 s' s_2 m + a_2 s s_1 m') \\ &= (\tfrac{1}{s_1 s \cdot s_2 s'} \otimes (a_1 s' s_2 m)) + (\tfrac{1}{s_1 s \cdot s_2 s'} \otimes (a_2 s s_1 m')) \\ &= (\tfrac{a_1 s' s_2}{s_1 s \cdot s_2 s'} \otimes m) + (\tfrac{a_2 s s_1}{s_1 s \cdot s_2 s'} \otimes m') \\ &= \tfrac{a_1}{s_1}(\tfrac{1}{s} \otimes m) + \tfrac{a_2}{s_2}(\tfrac{1}{s'} \otimes m') = \tfrac{a_1}{s_1} g(\tfrac{m}{s}) + \tfrac{a_2}{s_2} g(\tfrac{m'}{s'}). \end{aligned}$$

Finally, $\widehat{f}(g(\frac{m}{s})) = \widehat{f}(\frac{1}{s} \otimes m) = \frac{1}{s} \cdot \frac{m}{1}$ implies that $\widehat{f} \circ g = \mathrm{Id}_{S^{-1}M}$. Meanwhile, as $S^{-1}A \otimes M$ is generated by pure tensor elements it suffices to prove that $g(\widehat{f}(\frac{a}{s} \otimes m)) = \frac{a}{s} \otimes m$ for all pure tensor elements in order to prove that $g \circ \widehat{f} = \mathrm{Id}_{S^{-1}A \otimes_A M}$. Fortunately, $g(\widehat{f}(\frac{a}{s} \otimes m)) = g(\frac{a}{s} \cdot \frac{m}{1}) = \frac{1}{s} \otimes am = \frac{a}{s} \otimes m$.

Since $\widehat{f}$ is the inverse of an $S^{-1}A$-module homomorphism, we thus know that $\widehat{f}$ is $S^{-1}A$-linear. And now this proves that $S^{-1}A \otimes_A M \cong S^{-1}M$ as $S^{-1}A$-modules via an isomorphism identifying $\frac{a}{s} \otimes m$ with $\frac{am}{s}$.

To finish off, we claim that the isomorphisms described before induce a natural isomorphism between the tensor functor $T_{S^{-1}A} : A\text{-}\mathbf{mod} \to S^{-1}A\text{-}\mathbf{mod}$ and the localization by S functor. After all, given any A-module homomorphism $f : M \to M'$ we have that:

$$\begin{tikzcd}
\frac{a}{s}\otimes m \arrow[rrr, mapsto] \arrow[ddd, mapsto] & & & \frac{am}{s} \arrow[ddd, mapsto] \\
& S^{-1}A \otimes_A M \arrow[r, "\sim"] \arrow[d, "\mathrm{Id}_{S^{-1}A}\otimes f"] & S^{-1}M \arrow[d, "S^{-1}f"] & \\
& S^{-1}A \otimes_A M' \arrow[r, "\sim"'] & S^{-1}M' \arrow[r, equal] & \frac{f(am)}{s} \\
\frac{a}{s}\otimes f(m) \arrow[rrr, mapsto] & & & \frac{af(m)}{s}
\end{tikzcd}$$

Since localization by S is an exact functor, it follows $T_{S^{-1}A} : A\text{-}\mathbf{mod} \to S^{-1}A\text{-}\mathbf{mod}$ is an exact functor. Also note that the forgetful functor from $S^{-1}A$-**mod** to A-**mod** is an exact functor. Hence, we can conclude that the non-enriched tensor functor $T_{S^{-1}A} : A\text{-}\mathbf{mod} \to A\text{-}\mathbf{mod}$ is exact. This proves that $S^{-1}A$ is a flat A-module.

(d) Prove that $\frac{m_1 \otimes m_2}{1} \mapsto \frac{m_1}{1} \otimes \frac{m_2}{1}$ gives us a well-defined isomorphism:

$$S^{-1}(M_1 \otimes_A M_2) \xrightarrow{\sim} (S^{-1}M_1) \otimes_{S^{-1}A} (S^{-1}M_2).$$

Firstly, recall the existence of the forgetful functor $\mathcal{F} : S^{-1}A$-**mod** to A-**mod** which turns any $S^{-1}A$-module M into an A-module by defining $a \cdot m = \frac{a}{1}m$ and doesn't touch any of the morphisms. A fact I will need is that $S^{-1}\mathcal{F}(M) \cong M$ by the map sending $\frac{m}{s} \mapsto \frac{1}{s}m$.

(Well-defined):
Suppose $\frac{m}{s} = \frac{m'}{s'}$. Then there exists $t \in S$ such that $t(s'm - sm') = 0$. In turn, $\frac{1}{s}m - \frac{1}{s'}m' = \frac{ts'}{tss'}m - \frac{ts}{tss'}m' = \frac{1}{tss'}(t(s'm - sm')) = \frac{1}{tss'}(0) = 0$.

($S^{-1}A$-linear):
This is because
$\frac{a_1}{s_1}(\frac{1}{s}m) + \frac{a_2}{s_2}(\frac{1}{s'}m') = \frac{a_1 s_2 s'}{s_1 s s_2 s'}m + \frac{a_2 s_1 s}{s_1 s s_2 s'}m' = \frac{1}{s_1 s s_2 s'}(a_1 s_2 s' m + a_2 s_1 s m')$

(Injective):
Suppose $\frac{1}{s}m = 0$ in M. Then $m = \frac{s}{1} \cdot (\frac{1}{s}m) = \frac{s}{1} \cdot 0 = 0$. In turn, we have that $\frac{m}{s} = 0$ in $S^{-1}\mathcal{F}(M)$.

(Surjective):
For any $m \in M$ we have that $\frac{m}{1} \in S^{-1}\mathcal{F}(M)$ maps to m.

In particular, we have that $S^{-1}\mathcal{F}$ is naturally isomorphic to the identity functor on the category $S^{-1}A$-**mod**. After all, given any $S^{-1}A$-module homomorphism $f : M \to M'$ we have the following diagram:

$$\begin{tikzcd}
\frac{m}{s} \arrow[rrr, mapsto] \arrow[ddd, mapsto] & & & \frac{1}{s}m \arrow[ddd, mapsto] \\
& S^{-1}\mathcal{F}(M) \arrow[r, "\sim"] \arrow[d, "S^{-1}\mathcal{F}(f)"] & M \arrow[d, "f"] & \\
& S^{-1}\mathcal{F}(M') \arrow[r, "\sim"] & M' & \\
\frac{(\mathcal{F}(f))(m)}{s} = \frac{f(m)}{s} \arrow[rrr, mapsto] & & & \frac{1}{s}f(m)
\end{tikzcd}$$

Although I won't need this natural isomorphism until part (c).

Consider $(S^{-1}M_1) \otimes_{S^{-1}A} (S^{-1}M_2)$ as an A-module (by applying $\mathcal{F}$) and define:

$$f : M_1 \times M_2 \to (S^{-1}M_1) \otimes_{S^{-1}A} (S^{-1}M_2) \text{ by } f(m_1, m_2) = \frac{m_1}{1} \otimes \frac{m_2}{1}.$$

It's obvious that f is A-bilinear. Hence, by the universal property of the tensor product there exists a unique A-module homomorphism:

$$\widehat{f} : M_1 \otimes_A M_2 \to (S^{-1}M_1) \otimes_{S^{-1}A} (S^{-1}M_2)$$

with $\widehat{f}(m_1 \otimes m_2) = \frac{m_1}{1} \otimes \frac{m_2}{2}$. Upon localizing by S and using the identification described on the prior page, we thus get an $S^{-1}A$-module homomorphism

$$S^{-1}\widehat{f} : S^{-1}(M_1 \otimes_A M_2) \to (S^{-1}M_1) \otimes_{S^{-1}A} (S^{-1}M_2)$$

satisfying that $S^{-1}\widehat{f}(\frac{m_1 \otimes m_2}{1}) = \frac{m_1}{1} \otimes \frac{m_2}{2}$.

Going the other direction, define $g : S^{-1}M_1 \times S^{-1}M_2 \to S^{-1}(M_1 \otimes_A M_2)$ by $g(\frac{m_1}{s_1}, \frac{m_2}{s_2}) = \frac{m_1 \otimes m_2}{s_1 s_2}$. I claim that g is well-defined.

After all, suppose $\frac{m_i}{s_i} = \frac{m_i'}{s_i'}$ in $S^{-1}M_i$ for both i. Thus there exists $t_i \in S$ such that $t_i(s_i' m_i - s_i m_i') = 0$ in M_i for both i. In turn:

$$\begin{aligned}
\frac{m_1 \otimes m_2}{s_1 s_2} - \frac{m_1' \otimes m_2'}{s_1' s_2'} &= \frac{t_1 t_2 s_1' s_2' (m_1 \otimes m_2) - t_1 t_2 s_1 s_2 (m_1' \otimes m_2')}{t_1 t_2 s_1 s_1' s_2 s_2'} \\
&= \frac{(t_1 s_1' m_1 \otimes t_2 s_2' m_2) - (t_1 s_1 m_1' \otimes t_2 s_2 m_2')}{t_1 t_2 s_1 s_1' s_2 s_2'} = \frac{0}{t_1 t_2 s_1 s_1' s_2 s_2'} = 0
\end{aligned}$$

I also claim that g is $S^{-1}A$-bilinear.

The proof is analogous for linearity in either variable. So I'll just do the proof for the first variable of g.

- $g(\frac{a}{s} \cdot \frac{m_1}{s_1}, \frac{m_2}{s_2}) = \frac{(am_1) \otimes m_2}{s s_1 s_2} = \frac{a}{s} \cdot \frac{m_1 \otimes m_2}{s_1 s_2} = \frac{a}{s} g(\frac{m_1}{s_1}, \frac{m_2}{s_2})$.
- $g(\frac{m_1}{s_1} + \frac{m_1'}{s_1'}, \frac{m_2}{s_2}) = \frac{(s_1' m_1 + s_1 m_1') \otimes m_2}{s_1 s_1' s_2}$

$$\begin{aligned}
&= \frac{(s_1' m_1 \otimes m_2) + (s_1 m_1' \otimes m_1)}{s_1 s_1' s_2} = \frac{m_1 \otimes m_2}{s_1 s_2} + \frac{m_1' \otimes m_2}{s_1' s_2} \\
&= g(\frac{m_1}{s_1}, \frac{m_2}{s_2}) + g(\frac{m_1'}{s_1'}, \frac{m_2}{s_2})
\end{aligned}$$

By the universal property of tensor products, there is a unique $S^{-1}A$-module homomorphism $\widehat{g} : S^{-1}M_1 \otimes_{S^{-1}A} S^{-1}M_2 \to S^{-1}(M_1 \otimes_A M_2)$ satisfying that $\widehat{g}(\frac{m_1}{s_1} \otimes \frac{m_2}{s_2}) = \frac{m_1 \otimes m_2}{s_1 s_2}$.

To finish we claim that $S^{-1}\widehat{f}$ and $\widehat{g}$ are inverses. Because tensor product modules are generated by pure tensors, it suffices to show that $S^{-1}\widehat{f}(\widehat{g}(\frac{m_1}{s_1} \otimes \frac{m_2}{s_2})) = \frac{m_1}{s_1} \otimes \frac{m_2}{s_2}$ for all pure tensors $\frac{m_1}{s_1} \otimes \frac{m_2}{s_2}$ in order to prove that $S^{-1}\widehat{f} \circ \widehat{g} = \mathrm{Id}_{S^{-1}M_1 \otimes_{S^{-1}A} S^{-1}M_2}$.

Fortunately:

$$S^{-1}\widehat{f}(\widehat{g}(\tfrac{m_1}{s_1} \otimes \tfrac{m_2}{s_2})) = S^{-1}\widehat{f}(\tfrac{m_1 \otimes m_2}{s_1 s_2}) = \tfrac{1}{s_1 s_2} \cdot (\tfrac{m_1}{1} \otimes \tfrac{m_2}{1}) = \tfrac{m_1}{s_2} \otimes \tfrac{m_2}{s_2}.$$

Similarly, $S^{-1}(M_1 \otimes_A M_2)$ is generated by elements of the form $\frac{m_1 \otimes m_2}{1}$ where $m_1 \otimes m_2$ is a pure tensor in $M_1 \otimes_A M_2$. Hence, it suffices to show that $\widehat{g}(S^{-1}\widehat{f}(\frac{m_1 \otimes m_2}{1})) = \frac{m_1 \otimes m_2}{1}$ for all pure tensors $m_1 \otimes m_2$ in order to prove that $\widehat{g} \circ S^{-1}\widehat{f} = \mathrm{Id}_{S^{-1}(M_1 \otimes_A M_2)}$.

Fortunately $\widehat{g}(S^{-1}\widehat{f}(\frac{m_1 \otimes m_2}{1})) = \widehat{g}(\frac{m_1}{1} \otimes \frac{m_2}{1}) = \frac{m_1 \otimes m_2}{1}$.

(c) Prove that, if M is a flat A-module, then $S^{-1}M$ is a flat $S^{-1}A$-module.

Note that as M is flat, we know that T_M is an exact functor. In turn, if we denote the localization functor by S^{-1} and the forgetful functor by $\mathcal{F}$, we have that $S^{-1} \circ T_M \circ \mathcal{F}$ is an exact functor.

But now note that the isomorphisms from part (d) can be used to define a natural isomorphism between $S^{-1} \circ T_M \circ \mathcal{F}$ and $T_{S^{-1}M} \circ S^{-1} \circ \mathcal{F}$. After all, suppose $f : N \to N'$ is an $S^{-1}A$-module homomorphism. Then it suffices to check that the below diagram commutes for elements $\frac{m \otimes n}{1} \in S^1(M \otimes_A \mathcal{F}(N))$ since they form a generating set:

$$\begin{tikzcd}
\frac{m \otimes n}{1} \arrow[rrr, mapsto] \arrow[ddd, mapsto] & & & \frac{m}{1} \otimes \frac{n}{1} \arrow[ddd, mapsto] \\
& S^{-1}(M \otimes_A \mathcal{F}(N)) \arrow[r, "\sim"] \arrow[d, "S^{-1}(\mathrm{Id}_M \otimes f)"'] & (S^{-1}M) \otimes_{S^{-1}A} S^{-1}(\mathcal{F}(N)) \arrow[d, "\mathrm{Id}_{S^{-1}M} \otimes S^{-1}f"'] & \\
& S^{-1}(M \otimes_A \mathcal{F}(N')) \arrow[r, "\sim"] & (S^{-1}M) \otimes_{S^{-1}A} S^{-1}(\mathcal{F}(N')) & \\
\frac{m \otimes f(n)}{1} \arrow[rrr, mapsto] & & & \frac{m}{1} \otimes \frac{f(n)}{1}
\end{tikzcd}$$

Similarly, $T_{S^{-1}M} \circ S^{-1} \circ \mathcal{F} \cong T_{S^{-1}M}$ since we already showed $S^{-1} \circ \mathcal{F}$ is naturally isomorphic to the identity functor. And this proves that $T_{S^{-1}M}$ is an exact functor. So, $S^{-1}M$ is a flat $S^{-1}A$-module.

(e) Prove that if $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$-module for every $\mathfrak{p} \in \mathrm{Spec}(A)$, then M is flat.

Recall that M is flat if and only if $f : N \to N'$ being an injective A-module homomorphism implies $\mathrm{Id}_M \otimes f : M \otimes N \to M \otimes N'$ is an injective A-module homomorphism. Also by set 4 problem 2 (see *pages 623-624*), we have that $\mathrm{Id}_M \otimes f$ is injective if and only if $(\mathrm{Id}_M \otimes f)_{\mathfrak{p}}$ is injective for all $\mathfrak{p} \in \mathrm{Spec}(A)$.

Fortunately for any $\mathfrak{p} \in \operatorname{Spec}(A)$, if we let $S_{\mathfrak{p}}^{-1}$ denote the localization functor then the isomorphisms from part (d) induce a natural isomorphism between $S_{\mathfrak{p}}^{-1} \circ T_M$ and $T_{M_{\mathfrak{p}}} \circ S_{\mathfrak{p}}^{-1}$. In particular, this gives us the following commuting diagram:

$$\begin{tikzcd}
\frac{m\otimes n}{1} \arrow[rrr, mapsto] \arrow[ddd, mapsto] & & & \frac{m}{1}\otimes\frac{n}{1} \arrow[ddd, mapsto] \\
& (M\otimes_A N)_{\mathfrak{p}} \arrow[r, "\sim"] \arrow[d, "(\mathrm{Id}_M\otimes f)_{\mathfrak{p}}"] & M_{\mathfrak{p}}\otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \arrow[d, "\mathrm{Id}_{M_{\mathfrak{p}}}\otimes f_{\mathfrak{p}}"'] & \\
& (M\otimes_A N')_{\mathfrak{p}} \arrow[r, "\sim"'] & M_{\mathfrak{p}}\otimes_{A_{\mathfrak{p}}} N'_{\mathfrak{p}} & \\
\frac{m\otimes f(n)}{1} \arrow[rrr, mapsto] & & & \frac{m}{1}\otimes\frac{f(n)}{1}
\end{tikzcd}$$

But now as f is injective, we know that $f_{\mathfrak{p}}$ is also injective. Next, as $M_{\mathfrak{p}}$ is flat, we know that $\mathrm{Id}_{M_{\mathfrak{p}}} \otimes f_{\mathfrak{p}}$ is injective. And finally, by the above commuting diagram this means that $(\mathrm{Id}_M \otimes f)_{\mathfrak{p}}$ is injective. Since $\mathfrak{p}$ was an arbitrary prime ideal, this proves that $\mathrm{Id}_M \otimes f$ is injective. ■

Set 7 Problem 5: Suppose A is a unital commutative ring and K is a field.

(a) Suppose V and W are two K-vector spaces. Prove that $\dim_K(V \otimes_K W) = \dim_K(V) \cdot \dim_K(W)$

Firstly, as every K-vector space has a K-basis we can find isomorphisms $\theta : V \to \bigoplus_{i\in I} K$ and $\psi : W \to \bigoplus_{j\in J} K$. In turn, $\theta \otimes \psi$ is an isomorphism from $V \otimes_K W$ to $(\bigoplus_{i\in I} K) \otimes_K (\bigoplus_{j\in J} K)$.

We know it is an isomorphism since the homomorphism $\theta^{-1} \otimes \psi^{-1}$ acts as an inverse function on all pure tensor elements.

By set 6 problem 8 (see *pages 691-692*), we thus know that:

$$\begin{aligned} V \otimes_K W &\cong (\textstyle\bigoplus_{i\in I} K) \otimes_K (\bigoplus_{j\in J} K) \\ &\cong \textstyle\bigoplus_{j\in J}((\bigoplus_{i\in I} K) \otimes_K K) \\ &\cong \textstyle\bigoplus_{j\in J}(K \otimes_K (\bigoplus_{i\in I} K)) \cong \bigoplus_{j\in J}(\bigoplus_{i\in I}(K \otimes_K K)) \end{aligned}$$

Finally, $K \otimes_K K \cong K$ via the isomorphism sending $x \otimes y \to xy$ (see *page 699*). So, we get that:

$$V \otimes_K W \cong \bigoplus_{(i,j)\in I\times J} K \Longrightarrow \dim_K(V \otimes_K W) = |I| \cdot |J| = \dim_K(V) \cdot \dim_K(W)$$

(b) Suppose A is a local ring. Suppose M and N are finitely generated A-modules, and $M \otimes_A N = \{0\}$. Prove that either $M = 0$ or $N = 0$.

Suppose $\operatorname{Max}(A) = \{\mathfrak{m}\}$ and let $k := A/\mathfrak{m}$. Note that k is a field and an A-algebra. Hence, we can enrich the tensor functor T_k to be mapping from the category A-**mod** to the category k-**vect**.

Next, by the proposition on *page 699* we know that $k \otimes_A M = \frac{A}{\mathfrak{m}} \otimes_A M \cong \frac{M}{\mathfrak{m}M}$ as A-modules via the isomorphism sending $(a+\mathfrak{m}) \otimes x$ to $ax + \mathfrak{m}M$. Yet also recall from set 5 problem 3(b) (see *page 658*) that $\frac{M}{\mathfrak{m}M}$ is a well-defined k-vector space when we define $(a+\mathfrak{m})(x+\mathfrak{m}M) = ax + \mathfrak{m}M$. In particular, this means that

$$(b+\mathfrak{m}) \cdot ((a+\mathfrak{m}) \otimes x) = (ba+\mathfrak{m}) \otimes x \mapsto bax + \mathfrak{m}M = (b+\mathfrak{m}) \cdot (ax + \mathfrak{m}M)$$

And this proves that $k \otimes_A M \cong \frac{M}{\mathfrak{m}M}$ as k-vector spaces.

By analogous reasoning and isomorphisms, we have that $k \otimes_A N \cong \frac{N}{\mathfrak{m}N}$ as k-vector spaces.

By part (a) plus the fact that M and N are finitely generated (so that we can apply set 5 problem 3(b)), we know that:

$$d(M)d(N) = \dim_k(\tfrac{M}{\mathfrak{m}M}) \cdot \dim_k(\tfrac{N}{\mathfrak{m}N}) = \dim_k(\tfrac{M}{\mathfrak{m}M} \otimes_k \tfrac{N}{\mathfrak{m}N})$$

Yet now define a map $f : M \times N \to \frac{M}{\mathfrak{m}M} \otimes_k \frac{N}{\mathfrak{m}N}$ given by:

$$f(x,y) = (x+\mathfrak{m}M) \otimes (y+\mathfrak{m}N).$$

Here we view $\frac{M}{\mathfrak{m}M} \otimes_k \frac{N}{\mathfrak{m}N}$ as an A-module. As a reminder, if M' is an $\frac{A}{\mathfrak{m}}$-module then we can view M' as an A-module by setting $a \cdot x = (a+\mathfrak{m})x$ for all $a \in A$ and $x \in M'$.

This is easily checked to be A-bilinear. So, by the universal property of tensor products there exists a unique A-module homomorphism $\widehat{f} : M \otimes_A N \to \frac{M}{\mathfrak{m}M} \otimes_k \frac{N}{\mathfrak{m}N}$ satisfying that $\widehat{f}(x \otimes y) = (x+\mathfrak{m}M) \otimes (y+\mathfrak{m}N)$ for all $x \in M$ and $y \in N$. Yet as $M \otimes_A N = \{0\}$ and the image of $\widehat{f}$ is easily seen to include all pure tensors, this implies that all pure tensors of $\frac{M}{\mathfrak{m}M} \otimes_k \frac{N}{\mathfrak{m}N}$ are the zero element. And as they span the entire vector space, we have that $\dim_k(\frac{M}{\mathfrak{m}M} \otimes_k \frac{N}{\mathfrak{m}N}) = 0$. This proves that either $d(M) = 0$ or $d(N) = 0$, which in turn implies that either $M = \{0\}$ or $N = \{0\}$.

(c) Recall that $\operatorname{supp}(M) := \{\mathfrak{p} \in \operatorname{Spec}(A) : M_{\mathfrak{p}} \neq \{0\}\}$ (see *page 672*). Suppose that M and N are finitely generated A-modules. Prove that $\operatorname{supp}(M \otimes_A N) = \operatorname{supp}(M) \cap \operatorname{supp}(N)$.

By problem 4 of this problem set, we know that $(M \otimes_A N)_{\mathfrak{p}} \cong M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}}$. Also as $A_{\mathfrak{p}}$ is a local ring and $M_{\mathfrak{p}}$ and $N_{\mathfrak{p}}$ are finitely generated (since M and N are), we know from part (b) of this problem that $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} = \{0\}$ iff $M_{\mathfrak{p}} = \{0\}$ or $N_{\mathfrak{p}} = \{0\}$.

Therefore $(M \otimes_A N)_{\mathfrak{p}} = \{0\}$ iff $\mathfrak{p} \notin \operatorname{supp}(M)$ or $\mathfrak{p} \notin \operatorname{supp}(N)$. And this proves that $\operatorname{supp}(M \otimes_A N) = \operatorname{supp}(M) \cap \operatorname{supp}(N)$. ■